

Karl Pierce, PhD

Research Scientist



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LinkedIn

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Website

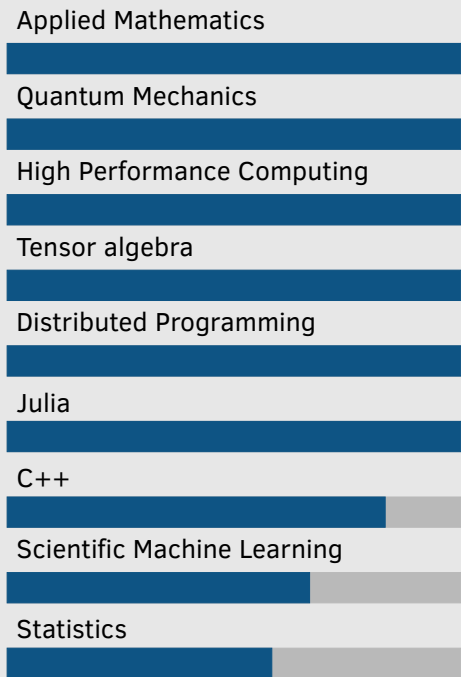


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Research Interests

- Linear and Multilinear Algebra
- High-dimensional Optimization
- High Performance Computing
- Library and Algorithm Design
- Randomized Approaches
- Sparse and Structured Tensors
- Electronic Structure Methods

Skills



Objective

Passionate about the application of emerging multilinear and randomized linear algebra techniques computational modeling problems motivated by physics and chemistry. Dedicated to creating advanced software tools to improve and accelerate the study of applied mathematics on modern distributed computer infrastructures.

Education

2021	PhD Theoretical Chemistry <i>Virginia Tech</i>	Blacksburg, VA
	Dissertation: Breaking the curse of dimensionality in electronic structure methods: towards optimal utilization of the canonical polyadic decomposition.	
2016	B.S. Chemical Physics <i>Rice University</i>	Houston, TX

Professional Experience

2025 - 2026	Postdoctoral Associate	The University of Maryland
	Postdoctoral Associate working under Dr. Haizhao Yang.	
	Projects:	
	- Introduction of tensor features to accelerate neural network training in scientific computing. - Interpolative QR acceleration of alternating least squares (ALS) optimization. - Random and adaptive pivot selection algorithms. - Block-wise sampling strategies to accelerate ALS. - Methods to accelerate the the finite expression (FEX) method for high-dimensional PDEs. - Tensor network and randomized optimization strategies for the butterfly matrix decomposition. - Reduced scaling quantum embedding algorithms. - Residual approximated coupled cluster methods.	
2022- 2025	Flatiron Software Research Fellow	The Flatiron Institute
	Software-focused postdoctoral researcher at the Center for Computational Quantum Physics.	
	Projects	
	- Developed an efficient pipeline for CPU/GPU agnostic algorithms in ITensors.jl. - Block-Sparse tensor contractions on GPU with NVIDIA's cuTENSOR team. - Mixed precision tensor network algorithms. - Approximated tensor contractions using gradient optimization for quantum physics and chemistry. - Developed ITensorCPD, a generic non-covariant algebra library for scientific computing.	

Publications

In Review

Accelerating the Canonical Polyadic Alternating Least Squares Optimization via a Randomized Interpolative Decomposition
Fakih, I; Grigori, L; [Pierce, K](#) Arxiv

On the Accelerated Learning of High Dimensional Functions with Tensor-Featured Networks
[Pierce, K](#); Khoo, Y, Yang, H

2025	Towards Using Matrix-Free Tensor Decomposition to Systematically Improve Approximate Tensor-Networks. Pierce, K J. Chem. Theory Comput. 2025, 21, 13, 6464–6481
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2026	STTID: High-Performance Sparse Tensor-Train Interpolative Decomposition. Meng, Z., Stoudenmire, M., Pierce, K., Mueller, F., Li, J. 2026 IEEE International Parallel and Distributed Processing Symposium (IPDPS), 15–31.
2026	A CPD-enabled low-scaling environment solver in a coupled cluster based static quantum embedding theory <u>Pierce, K</u> ; Aziz, M; Shee, A; FM Faulstich; (2026) The Journal of Chemical Physics, 164(21).
2025	Using Matrix-Free Tensor-Network Optimizations to Construct a Reduced-Scaling and Robust Second-Order Møller-Plesset Theory. <u>Pierce, K</u> ; Morales, M J. Chem. Theory Comput. 2025, 21, 12, 5952–5964
2022	Efficient Construction of Canonical Polyadic approximations of Tensor Networks <u>Pierce, K</u> ; Valeev, E. J. Chem. Theory Comput. 2023, 19, 1, 71–81.
2021	Robust Approximation of Tensor Networks: Application to Grid-Free Tensor Factorization of the Coulomb Interaction. <u>Pierce, K</u> ; Rishi, V; Valeev, E. F. J. Chem. Theory Comput. 2021, 17, 4, 2217–2230.
2020	Massively Parallel Quantum Chemistry: A high-performance research platform for electronic structure. Peng C; Lewis C; Xiao W; Clement M; <u>Pierce, K</u> ; Rishi V; Pavošević F; Slattery S; Zhang J; Teke N; Kumar A; Masteran C; Asadchev A; Calvin J; Valeev E. F; (2020). J. Chem. Phys., 153(4), 44120.
Accepted	TIDES: Tiered Block-Sparse Tensor Contraction with Streaming Transpose on GPUs Li, Z; Deverapalli, S; Zhao, Y; Stoudenmire E.M.; Mueller, F; <u>Pierce, K</u> ; Li, J. Processing for the International Conference for High Performance Computing, Networking, Storage, and Analysis.
In Preparation	Weighted Adaptive Random Pivoting for Interpolative and CUR Decompositions. Pearce, K; <u>Pierce, K</u> Leveraging Tree Operators to Accelerate the Finite Expression Optimization of High Dimensional PDE <u>Pierce, K</u> ; Yi, C; Sun, Y; Yang, H Blocked Leverage Scores in Randomized Least Squares for CP Tensor Decomposition <u>Pierce, K</u> ; Bellavita, J; Liu, Y; Nahar Keya, K; Haddock, J; Kolda, T Studying the Efficient Approximation of Lattice Networks using the Canonical Polyadic Decomposition <u>Pierce, K</u> ; Tindall, J; Stoudenmire EM Tensor Hypercontraction, Residual Approximated Rank-Reduced Coupled Cluster, I; Molecular Energies and Energy-based Properties. Pimpel, F; Morales, M; Grüneis, A; <u>Pierce, K</u> GPU Performance of Dense and Sparse Tensor Algorithms using the ITensor Library <u>Pierce, K</u> ; Fishman, M; EM Stoudenmire An Efficient Application of the Canonical Polyadic Decomposition to the Rank-Reduced Coupled Cluster Method <u>Pierce, K</u> Effective use of 4-way Canonical Polyadic Decomposition for Accelerating the Coupled-Cluster Perturbative Triples. <u>Pierce, K</u> ; Bao, Y; Pavošević, F; Valeev, E

Invited Presentations and Posters

2027	Joint Mathematics Meetings of the American Mathematical Society Predicting Coupled Cluster Wavefunction Amplitude Tensors via an Equivariant Graph Neural Network	Seminar
2026	Workshop: Tensor Decompositions in Polynomially-scaling Many-body Expansion Methods The Canonical Polyadic Decomposition of Two-Electron Integrals in Quantum Chemistry	Seminar

2026	Invited Seminar at Rutgers University In the Pursuit of Low-Scaling Electronic Structure Methods via Tensor-based Approaches and Approximations.	Seminar
2026	RandNLA Workshop Investigating Efficient Leverage Score Sampling Methods for the Alternating Least Squares Optimization of the Canonical Polyadic Decomposition	Poster
2026	27th Conference of the International Linear Algebra Society Blocked Leverage Score Sampling in the Randomized Least Squares for CP Tensor Decomposition	Seminar
2026	SIAM Conference on Parallel Processing for Scientific Computing Accelerating the Canonical Polyadic Decomposition via a Randomized Interpolative Sampling Procedure	Seminar
2025	Molecular Quantum Mechanics Conference Application of the Canonical Polyadic Decomposition in the Pursuit of Extremely Low-Scaling Correlation Methods.	Seminar
2025	NVIDIA Design, Demos, Ideas, and Research Discussions Studying the effective application of low-rank tensor approximations to electronic structure methods.	Seminar
2025	SIAM Conference on Computational Science and Engineering Towards the Efficient Approximation of Higher-Order, Tensor-Network Contractions Via a Low-Rank, Matrix-Free Tensor Decomposition	Seminar
2025	Applied Math Colloquium at Rensselaer Polytechnic Institute Investigating the Utility of Matrix-Free Tensor Decompositions in Quantum Simulations.	Seminar
2024	Juliacon Improving the life-cycle of tensor algorithm development	Seminar
2024	Invited group meeting: Virginia Tech Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2024	Extreme-scale Mathematically-based Computational Chemistry meeting Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2024	Invited group meeting: Vienna University of Technology Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2023	SIAM Conference on Computational Science and Engineering Introduction to the ITensor Software Library for Tensor Network Calculations	Seminar
2021	Colloquium at Vienna University of Technology <i>Utility of the Canonical Polyadic Decomposition and Robust Tensor Network Approximations</i>	Seminar
2019	Virginia Tech Department of Chemistry Internal Seminar <i>Reduced Cost Electronic Structure Theory via the Canonical Polyadic Decomposition</i>	Seminar
2019	American Chemical Society National Meeting <i>Towards Reduced Scaling Higher Order Coupled Cluster Methods via Tensor Decomposition.</i>	Poster
2018	Modern Wavefunction Methods in Electronic Structure Theory <i>Reducing Complexity and Cost of High-Order Coupled-Cluster Method via Canonical Polyadic Decomposition of Hamiltonian</i>	Poster
2018	Penn Conference in Theoretical Chemistry and Electronic Structure Workshop <i>Reducing Complexity and Cost of High-Order Coupled-Cluster Method via Canonical Polyadic Decomposition of Hamiltonian.</i>	Poster
2018	Virginia Tech Department of Chemistry Preliminary Exam <i>Reduced Scaling of Accurate Electronic Structure Methods using Tensor Decompositions</i>	Seminar
2017	Southeast Theoretical Chemistry Association Meeting <i>Toward Efficient Canonical Product Decomposition in TiledArray Framework</i>	Poster

Workshops and Advanced Topic Schools

2026	RandNLA workshop Seminar-focused workshop to discuss the current direction of randomized linear algebra and to develop collaborations in the field.	Lausanne, Switzerland
2025	Research Collaboration Workshop, "Randomized Numerical Linear Algebra" (RNLA) Research Workshop that focuses on developing efficient and practical algorithms for solving problems using numerical linear algebra via randomized techniques. Assigned to work with Dr. Tamara Kolda on a project to study structure-aware randomization for linear algebra.	Los Angeles, CA
2022	Argonne Training Program on Extreme-scale Computing Participated in an intensive two week summer school learning modern key skills, approaches, and about tools to design, implement and execute scientific applications on state of the art, leadership-class computing systems of today and the future.	Chicago, IL
2018	MolSSI Summer School and Workshop Parallel Computing in Molecular Sciences Participated in a three-day lecture series where researchers in academia and from Berkeley national lab. Discussed computational parallelism and communication on homogenous and heterogeneous CPU/GPU computer systems.	Berkely, CA
2018	Modern Wavefunction Methods in Electronic Structure Theory Attended a week-long summer school at the Max-Planck institute in Germany directed towards Ph.D. students and postdocs with aims to teach advanced topics in the field of ab initio electronic structure theory, reduced scaling algorithms, and software implementations on modern hardware.	Gelsenkirchen, Germany

Awards

2026	Early Career NSF Conference Grant ILAS 2026 conference travel grant	International Linear Algebra Society
2021	Graduate School Doctoral Assistanship Award Award for excellence in research in leadership	Virginia Tech
2012,2014	C-USA Commissioner's Honor Roll	Rice University

Service

2027	Organized Tensor Algebra Minisymposium at SIAM CSE
2025-2027	Advisory Committee Member for NSF Cross Project
2025	Panel Member for Society for Industrial and Applied Mathematics undergraduate scholars

Certificates

2026	LinkedIn Statistics Foundations 1: The Basics
2025	LinkedIn Training Neural Networks in Python
2022	NVIDIA Certificate in Scaling CUDA C++ Applications to Multiple Nodes
2022	NVIDIA Certificate in Fundamentals of Accelerated Computing with CUDA C/C++

Teaching and Mentoring Experience

Course Work

University of Maryland DATA 350: Data Visualization and Presentation

The Flatiron Institute

[Graduate Scholar Workshop Series](#)

[qnumerics](#): School for Numerical Methods in Quantum Information Science

Virginia Tech

Physical Chemistry, Thermodynamics
General Chemistry Lab
Physical Chemistry Lab
Virginia Tech: General Chemistry Lab
General Tutoring

Mentoring

Mina Mandic	Flatiron Institute Undergraduate Summer internship.
Florian Pimpel	PhD Candidate at TU Wien. Studying reduced-scaling approximations for periodic coupled cluster.
Muhammad Aziz	PhD Student at Rensselaer Polytechnic Institute. Studying reduced-scaling approximations for chemistry-embedding problems.
Israa Fakih	PhD Student at the Institute of Mathematics, EPFL. Studying the application of randomized linear algebra in tensor decompositions algorithms.
Kamrun Nahar Keya	PhD Student at the Arizona State University. Studying the canonical polyadic decomposition and computer programming fundamentals.

Software Projects

ITensorCPD.jl	Developed an efficient tensor algebra library to support the canonical polyadic decomposition of tensors and tensor networks interfacing with ITensors.jl and ITensorNetworks.jl. Implementing novel algorithms using randomized linear algebra.
ITensors.jl	Developed generic libraries to seamlessly integrate GPU support. Worked with lead developers to reorganize, simplify and extend the functionality of the ITensor and internal Libraries.
BTAS	Developed efficient serial tensor algebra kernels. Implemented existing and novel tensor decomposition algorithms.
TiledArray	Developed distributed tensor algebra kernels using the OMP based task scheduling software madness . Developed an efficient block distributed canonical polyadic decomposition.
MPQC	Developed canonical electronic structure methods such as Hartree-Fock, Møller-Plesset 2 perturbation theory, and various levels of coupled cluster theory. Utilized higher-order decompositions schemes to develop reduced-cost and scaling implementations of canonical electronic structure methods.
MPCC	Developing a python-based library interface to tackle large-scale quantum chemistry embedding problems by effectively combining existing high performance interfaces using generic programming strategies. To be released soon.

Professional Affiliations

The American Chemical Society
Society of Industrial and Applied Mathematicians

Extra-curricular Activities

2017-2023	Pole Vault Coach	Blacksburg High School
	Designed individualized athletic training and programming as a head coach for youth athletes.	
2012-2016	Division 1 Athlete	Rice University
	Participated in Division 1 athletics at Rice University as a pole vaulter on the track and field team.	